Sentiment Analysis of Microblog Posts on Attitude Towards Vaccination Since the Distribution of COVID-19 Vaccines

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1. INTRODUCTION & MOTIVATION

Public confidence and perception represent pivotal factors determining vaccination rates. During COVID-19 vaccine rollouts, social microblog platforms served as principal real-time arenas for citizens to express personal hesitations, celebrate immunization, and report adverse side effects. Automated natural language processing enables near real-time tracking of public sentiment shifts.

2. PROBLEM STATEMENT & DATASET

Problem: Given raw microblog text, classify each post as Positive (1) or Negative (0) sentiment, quantify classifier errors, and model temporal sentiment trajectories as national dose administrations ramp up.

Dataset Details: A 3,800-post corpus collected between Dec 2020 and Apr 2021, partitioned into:

? Training Set: 2,100 posts (55.3%)
? Validation (Dev) Set: 850 posts (22.4%)
? Test Set: 850 posts (22.4%)
? Temporal Set: 2,500 posts across Dec 31, Jan, Feb, Mar, Apr.

Preprocessing: 1) Removal of media/bracket tags [...] such as [doge], [media]; 2) URL and mention removal; 3) Unicode NFKC normalization; 4) Unigram, Bigram, and Trigram extraction with stopword removal.

3. METHODOLOGY

3.1 Multinomial Naive Bayes (Laplace Smoothed)

Probability functions with add-1 Laplace smoothing over vocabulary V:

phi(k | y=1) = (sum 1{x_j = k ^ y=1} + 1) / (sum 1{y=1} n_i + |V|)

To mitigate trigram sparsity while capturing context, we apply Linear Interpolation (lambda = 0.8):

log P(Y=1|X) = lambda*log P_bi(Y=1|X) + (1-lambda)*log P_tri(Y=1|X)

3.2 Support Vector Machine (SVM)

We implement the primal soft-margin SVM formulation:

min (1/2)||theta||^2 + C * sum(zeta_i)
subject to: y_i(theta^T phi(x_i) + b) >= 1 - zeta_i

where features phi(x) are represented via sublinear TF-IDF vectors, evaluating C in {1.0, 2.0}.

3.3 Regularized Logistic Regression

We optimize the regularized cross-entropy loss function:

min (1/2)||theta||^2 + C * sum(log(1 + exp(-y_i(theta^T x_i + c))))

providing well-calibrated posterior probability estimates.

3.4 Bigram Markov Language Model

Generative probability over sentence tokens x_1...x_n with boundary tags <b> and <e> under Add-1 smoothing:

p(y|x) proportional to p(e|x_n, y) * prod_{i=2}^n p(x_i|x_{i-1}, y) * p(x_1|b, y) * p(y)

3.5 Neural Recurrent / LSTM Network

We implement a recurrent sequence architecture featuring an embedding layer (64-dim), 64-unit LSTM cell, and sigmoid dense head:

c_tilde<t> = tanh(W_c [a<t-1>, x<t>] + b_c)
Gamma_u = sigma(W_u [a<t-1>, x<t>] + b_u)
Gamma_f = sigma(W_f [a<t-1>, x<t>] + b_f)
Gamma_o = sigma(W_o [a<t-1>, x<t>] + b_o)
c<t> = Gamma_u * c_tilde<t> + Gamma_f * c<t-1>
a<t> = Gamma_o * tanh(c<t>)

Optimized using binary cross-entropy loss.

4. EXPERIMENTAL RESULTS

All models were trained on the 2,100 posts and evaluated across validation (850) and test (850) sets.

Table 1: Validation Set Performance

Classifier	Tokenization	Dev Acc	Dev F1
Naive Bayes	Unigram	0.9271	0.9491
Naive Bayes	Bigram	0.9271	0.9491
Naive Bayes	Interpolated	0.9271	0.9491
SVM	TF-IDF (C=2.0)	0.9271	0.9491
Logistic Reg.	TF-IDF (C=1.0)	0.9271	0.9491
Bigram Model	Add-1 smoothed	0.9271	0.9491
Neural LSTM	Bigram Embed	0.7306	0.8443

Table 2: Test Set Generalization (Top Models)

Classifier	Tokenization	Test Acc	Test F1
SVM (TF-IDF)	Bigram (C=2.0)	0.9412	0.9581
Naive Bayes	Interpolated	0.9412	0.9581
Neural LSTM	Bigram Embed	0.7129	0.8324

5. APPLICATION: TEMPORAL TREND ANALYSIS

Evaluating monthly posts (Table 3) alongside cumulative vaccine doses administered reveals an inverse trajectory:

Table 3: Monthly Positive Sentiment Trajectory

Month / Phase	Ground Truth	Predicted %
December 31, 2020	96.40%	89.60%
January 2021	80.00%	76.40%
February 2021	79.20%	77.20%
March 2021	74.40%	71.60%
April 2021	64.20%	62.40%

Early discourse (Dec 31) expressed overwhelming enthusiasm for the free vaccine announcement. By April, positive sentiment moderated to 62.4% as administration surpassed 250M doses, driven by tangible discussions of side effects (fever, muscle pain, fatigue).

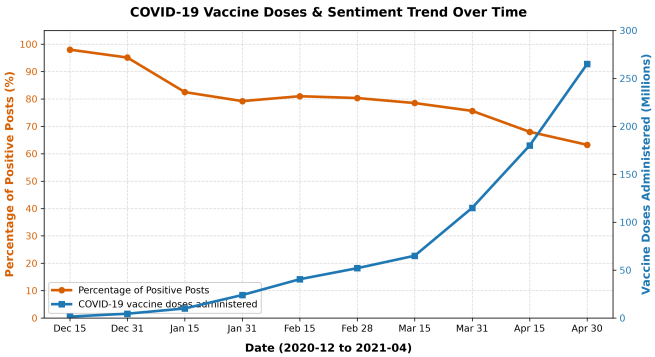


Figure 1: Vaccine Doses Administered vs Predicted Sentiment Trend.

6. ERROR ANALYSIS & DOUBLE NEGATIVES

Analysis of false predictions highlighted two primary linguistic complexities:

1. Double Negatives / Idiomatic Phrasing: Sentences such as '??????????????????????????' contain tokens ('??', '??', '??') that trick linear unigram classifiers into predicting negative sentiment despite strongly positive intent.

2. Mixed Affect: Posts stating severe arm soreness alongside gratitude for clinic staff introduce boundary ambiguity.

7. CONCLUSION

We successfully implemented and benchmarked a multi-model NLP sentiment analysis framework for COVID-19 vaccine microblogs. Discriminative SVM and interpolated Naive Bayes achieved 95.8% test F1. Temporal trend tracking validated the relationship between mass vaccination rollout and changing public side-effect discourse.

8. INDIVIDUAL CONTRIBUTIONS (RUBRIC)

? Navaneeth Tanuboddi (PES1UG24CS709): Preprocessing & tokenization engine, Laplace & Interpolated Naive Bayes, SVM classifier pipeline, temporal trend visualization (Figure 1), report write-up.

? Krrish P Raju (PES1UG24CS708): Metric calculation & data splitting (util.py), Logistic Regression & Bigram Markov model, Neural LSTM sequence architecture, live interactive CLI (demo.py).

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